

Homework #1 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

Given position and velocity of a spacecraft in an inertial reference frame and an epoch vector, estimate position and velocity for each epoch and graph the trajectory of the spacecraft in the inertial reference frame. Using the data from 15/01/1995 @ 00:00:00 in Table 1, graph the trajectory over the next 24 hours.

GPS	r_0 [km]	13426.726354	-20645.023752	10108.071628
	v_0 [km/s]	2.638622	0.329454	-2.805018
Molniya	r_0 [km]	2869.944136	-1656.963019	-6617.757380
	v_0 [km/s]	4.814217	8.338469	0.000000
Geosynchronous	r_0 [km]	-29260.792529	23865.119693	23865.119693
	v_0 [km/s]	-2.334835	-1.212853	-1.212853

TABLE 1. Initial conditions

Assuming a spherical Earth rotating at a constant angular velocity, estimate and graph the ground track of a spacecraft, whose position is known in the inertial reference frame for given moments of time. Using the data from 15/01/1995 @ 00:00:00 in Table 1, graph the ground track over the next 24 hours.

1.2 Calculation of classical orbital elements

Given $\vec{r}_0$ and $\vec{v}_0$ in an inertial reference frame it's possible to calculate the classical orbital elements COE = $\{a, e, i, \omega, \Omega, \nu\}$. Five orbital elements are necessary to describe the motion of a spacecraft in orbit, while the parameter ν defines its position at a given time.

While the first two elements define respectively the size and shape of the conic in the orbit plane, the last three determine the tilt of the spacecraft's orbit around Earth, the angle from the spacecraft's ascending node to its periapsis, measured in the direction of motion and at last the angle from a fixed versor $\hat{\gamma}$, the vernal equinox, to the direction of the ascending node, measured in a equatorial plane. Values are shown in Table 2.

$$\begin{aligned}h &= |\vec{h}| = |\vec{r}_0 \wedge \vec{v}_0| \\e &= |\vec{e}| = \left| \frac{\vec{v}_0 \wedge \vec{h}}{\mu} - \frac{\vec{r}_0}{r_0} \right| \\a &= \frac{h^2}{\mu(1 - e^2)}\end{aligned}$$

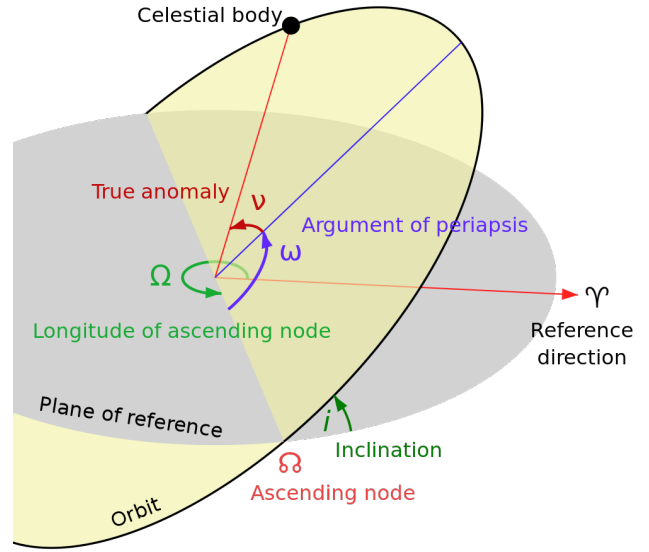
$$i = \arccos(\hat{g}_3 \cdot \hat{h})$$

$$\vec{n} = \hat{g}_3 \wedge \vec{h}$$

$$\Omega = \begin{cases} \arccos(\hat{g}_1 \cdot \hat{n}) & \text{if } n_2 > 0 \\ 2\pi - \arccos(\hat{g}_1 \cdot \hat{n}) & \text{if } n_2 < 0 \end{cases}$$

$$\omega = \begin{cases} \arccos(\hat{n} \cdot \hat{e}) & \text{if } e_3 > 0 \\ 2\pi - \arccos(\hat{n} \cdot \hat{e}) & \text{if } e_3 < 0 \end{cases}$$

$$\nu_0 = \begin{cases} \arccos(\hat{e} \cdot \hat{r}_0) & \text{if } \vec{r}_0 \cdot \vec{v}_0 > 0 \\ 2\pi - \arccos(\hat{e} \cdot \hat{r}_0) & \text{if } \vec{r}_0 \cdot \vec{v}_0 < 0 \end{cases}$$



Orbit	a [km]	e	i [deg]	ω [deg]	Ω [deg]	ν_0 [deg]
GPS	2.6560e+04	0.0035	55.0804	21.5032	139.6893	130.9108
Molniya	2.6562e+04	0.7214	63.4000	270.0000	60.0000	360.0000
Geosynchronous	4.2164e+04	0.1000	45.0000	6.1038e-04	0	130.9239

TABLE 2. Orbital elements

1.3 Time Problem

Once all orbital elements are found, the orbital elements can be propagated and thus updated along the orbit to draw the spacecraft's trajectory. In order to do so, one must resolve the time problem with some help from *Kepler's equation* to pass from Δt to $\Delta \nu$. We've used a time interval that goes from 15/01/1995 @ 00:00:00 to 16/01/1995 @ 00:00:00. To simplify calculations Modified Julian Date (MJD) was used throughout the homework.

$$\text{MJD} = \text{JD} - 2400000.5$$

$$t_{\text{span}} = [49732, \dots, 49733] \times T_{\text{solare}}$$

For each interval we'll have to go through a series of operations to transform the true anomaly in eccentric anomaly, to invert *Kepler's equation* using the *Newton-Raphson method* and to take that value back to true anomaly.

$$\Delta t = t_{\text{span}}(j) - t_{\text{span}}(1)$$

$$E_j = 2 \arctan \left(\sqrt{\frac{1-e}{1+e}} \tan \frac{\nu_j}{2} \right)$$

$$M_j = E_j - e \sin E_j$$

$$\Delta M = \sqrt{\frac{\mu}{a^3}} \Delta t$$

$$\begin{aligned}
 M_{j+1} &= M_j + \Delta M \\
 E_{j+1} &= E_{j+1,k+1} \\
 \nu_{j+1} &= 2 \arctan \left(\sqrt{\frac{1+e}{1-e}} \tan \frac{E_{j+1}}{2} \right)
 \end{aligned}$$

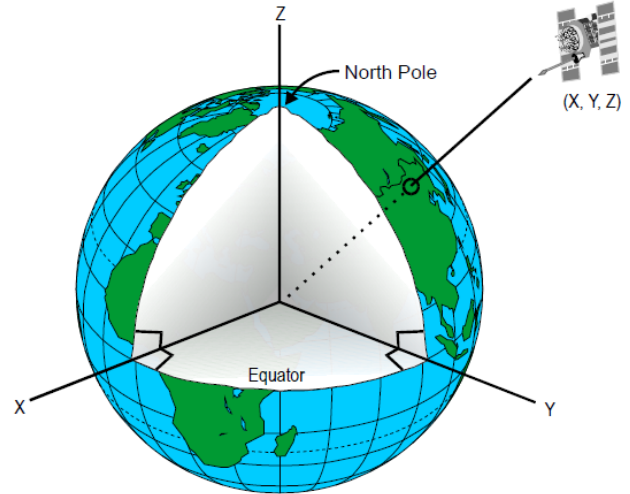
where we used the following equations to invert *Kepler's equation*

$$\begin{aligned}
 E_{j+1,0} &= M_{j+1} \\
 E_{j+1,k+1} &= E_{j+1,k} - \frac{f(E_{j+1,k})}{f'(E_{j+1,k})} = E_{j+1,k} - \frac{E_{j+1,k} - e \sin E_{j+1,k} - M_{j+1}}{1 - e \cos E_{j+1,k}}
 \end{aligned}$$

1.4 Perifocal-ECI rotation

Both the Perifocal and Earth Centered Inertial (ECI) frames are valid to analyze the orbit motion as both have their origins at the center of mass of Earth and do not rotate with respect to the stars. The former having unit vectors $\{\hat{e}, \hat{p}, \hat{h}\}$ is placed on top of the orbital plane, granting either position $\vec{r}$ or velocity $\vec{v}$ no component along the normal to the plane. Also, it uses the eccentricity direction, thus meaning the true anomaly ν can be used to describe the motion. This ensures a simplified way to write the position and velocity expressions.

$$\begin{aligned}
 \vec{r}|_{PER} &= \frac{p}{1 + e \cos \nu} \begin{bmatrix} \cos \nu \\ \sin \nu \\ 0 \end{bmatrix} \\
 \vec{v}|_{PER} &= \sqrt{\frac{\mu}{p}} \begin{bmatrix} e \sin \nu \\ 1 + e \cos \nu \\ 0 \end{bmatrix}
 \end{aligned}$$



Instead, the latter is placed on top of the equatorial plane, with its first axis along the vernal equinox $\hat{\gamma}$. The Sun lies in the direction of the vernal equinox around 21 March. Also, the third axis is set along the Earth rotation axis $\hat{z}$. For objects in space, the equations of motion that describe orbital motion are simpler in a non-rotating frame such as ECI.

The rotation matrix can be found using the angles ω , i and Ω as a product of three simple rotations, although their order is significant. Notably a first rotation of value Ω is applied on the third axis, followed by a second rotation of value i around the first axis and at the end a third rotation of value ω around the third axis.

$$\begin{aligned}
 R_{ECI \rightarrow PER} &= R_3(\omega) R_1(i) R_3(\Omega) = \begin{bmatrix} \cos \omega & \sin \omega & 0 \\ -\sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix} \begin{bmatrix} \cos \Omega & \sin \Omega & 0 \\ -\sin \Omega & \cos \Omega & 0 \\ 0 & 0 & 1 \end{bmatrix} = \\
 &= \begin{bmatrix} \cos \Omega \cos \omega - \sin \Omega \sin \omega \cos i & \sin \Omega \cos \omega + \cos \Omega \sin \omega \cos i & \sin \omega \sin i \\ -\cos \Omega \sin \omega - \sin \Omega \cos \omega \cos i & -\sin \Omega \sin \omega + \cos \Omega \cos \omega \cos i & \cos \omega \sin i \\ \sin \Omega \sin i & -\cos \Omega \sin i & \cos i \end{bmatrix}
 \end{aligned}$$

$$R_{PER \rightarrow ECI} = [R_{ECI \rightarrow PER}]^T$$

Found the rotation matrix, the position and velocity vectors can be evaluated in the ECI frame. Finally we can graph the 3D trajectory using these vectors.

$$\begin{aligned}\vec{r}|_{ECI} &= R_{PER \rightarrow ECI} \vec{r}|_{PER} \\ \vec{v}|_{ECI} &= R_{PER \rightarrow ECI} \vec{v}|_{PER}\end{aligned}$$

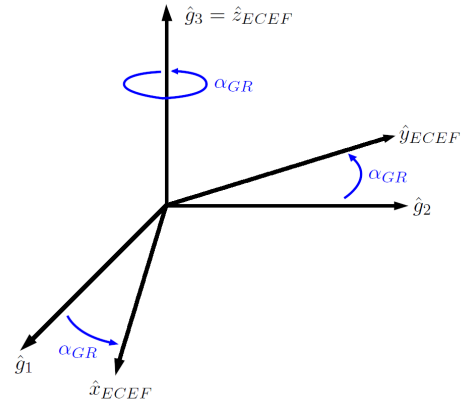
1.5 The ECEF frame

The ECEF frame is a non inertial, Earth fixed frame of reference. Its axes are fixed with respect to Earth's surface, with the z axis being the rotational axis. A point on the surface ($r = R_{\oplus}$) is defined in this reference frame using latitude and longitude. The ground track of an orbiting spacecraft can be correctly drawn using ECEF, if for each point of the trajectory the latitude and longitude of the subsatellite point are found. To be able to relate this frame with the inertial ECI, we need a meridian serving as a reference, the Greenwich meridian. By knowing where it's positioned at a certain time ($\alpha_{GR} = 100^\circ$ @ 01/01/95) we can estimate its future positions using the following equation. Remember that $\omega_{\oplus} = \frac{2\pi}{T_{sid}}$.

$$\alpha_{GR}(t) = \alpha_{GR}(t_0) + \omega_{\oplus}(t - t_0)$$

$$R_{ECI \rightarrow ECEF} = R_3(\alpha_{GR}) = \begin{bmatrix} \cos \alpha_{GR} & \sin \alpha_{GR} & 0 \\ -\sin \alpha_{GR} & \cos \alpha_{GR} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{r}|_{ECEF} = R_{ECI \rightarrow ECEF} \vec{r}|_{ECI}$$

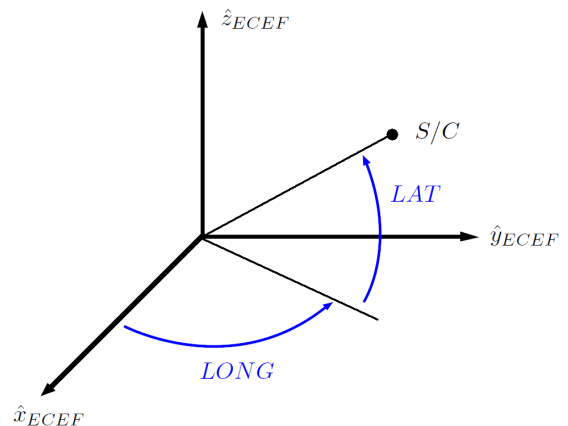


1.6 Calculation of latitude and longitude

Lastly, we need to calculate the latitude and longitude of the subsatellite point along the entire trajectory using some trigonometric functions. Only then we'll be able to graph the ground track.

$$LAT = \arcsin \left(\frac{r_{ECEF}(3)}{|r_{ECEF}|} \right)$$

$$LONG = \text{atan2} (r_{ECEF}(2), r_{ECEF}(1))$$



1.7 Trajectory and ground track representation

- **GPS orbit**

The *GPS orbit* is a special orbit suited for a radio-navigation service known as the Global Positioning System, or GPS. These transmissions are controlled by highly stable atomic clocks on board the satellites. Operated by the US military, it comprises a constellation of 24 to 32 satellites scattered around 6 different orbit planes to ensure that the users will have, at least, 8 simultaneous satellites in view from any point at the Earth surface at any time. Each orbital plane carries 4 satellites shifted along the orbit. Each satellite broadcasts radio waves towards Earth that contain information regarding its position and time. By combining signals transmitted by several satellites and received simultaneously, a GPS receiver can calculate its position on the Earth with an accuracy of approximately 10 m.

Orbits are nearly circular, with eccentricity less than 0.02, a semi-major axis of 26560 km. This puts them in the medium Earth orbit (MEO) range. The satellites have a nominal period of 12 h sidereal time and complete two full orbits every day.

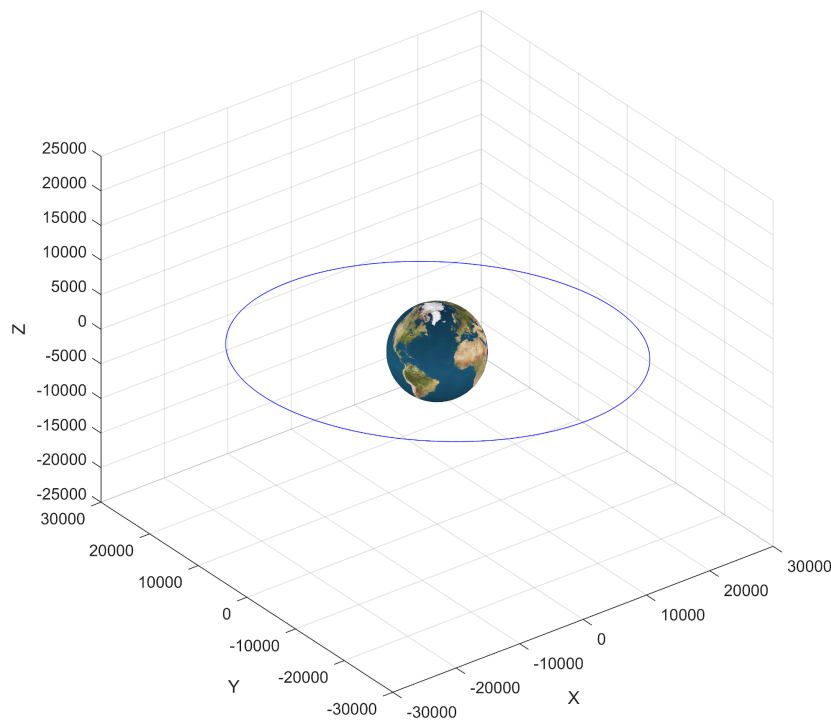


FIGURE 1. GPS orbit trajectory

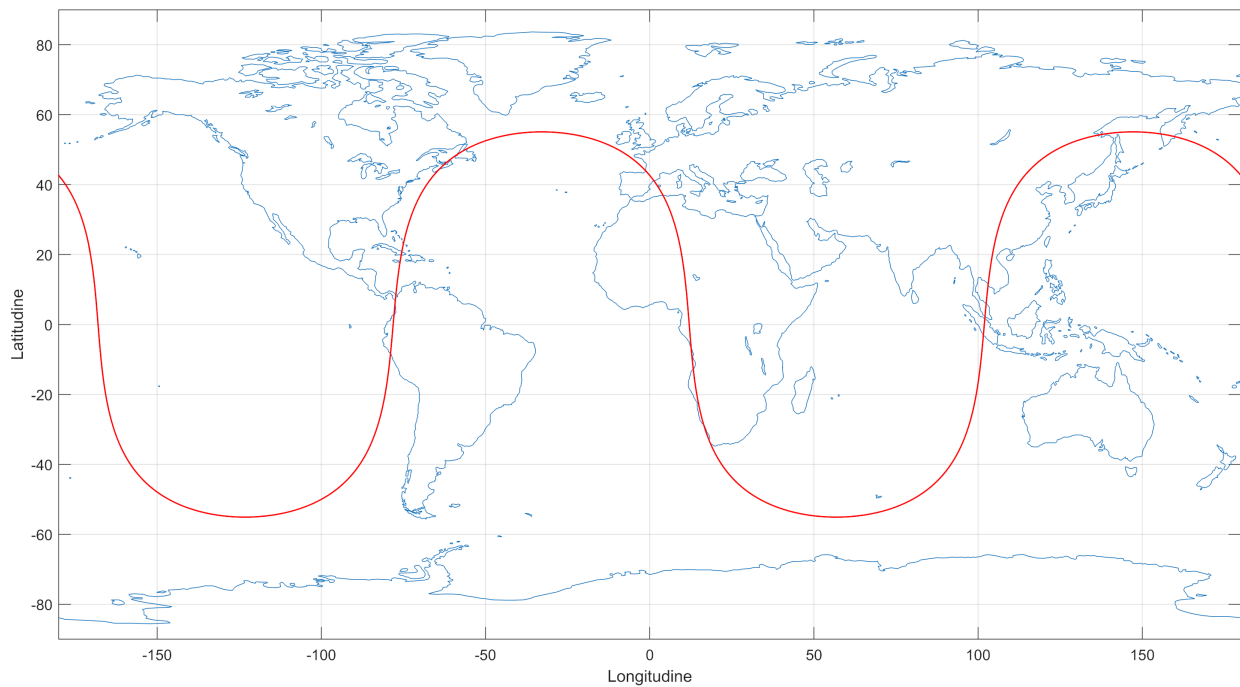


FIGURE 2. GPS orbit ground track

- **Molniya orbit**

The *Molniya orbit* is a frozen orbit that allows for communications and remote sensing coverage over high latitudes.

Discovered for the first time by Soviet scientists in the 1960s as an alternative to geostationary orbits, which would have required higher Δv when launched even from USSR's southernmost spaceport, Baikonur.

The drawbacks are that the ground station need a steerable antenna for communications, the spacecraft will pass the Van Allen radiation belt four times per day and 3 satellites are needed to have continuous coverage.

In the Keplerian model we considered Earth to have an uniform and spherical distribution of masses, however for a deeper analysis of the spacecraft motion we need to consider the generated asymmetrical gravity field. The most important effect, also called J_2 is due to more mass being concentrated over the equator. Because of this perturbing effect some parameters (such as Ω and ω undergo a secular variation).

$$\dot{\omega} = \frac{3}{2}n_0J_2\left(\frac{r_{\oplus}}{p}\right)^2\left(2 - \frac{5}{2}\cos^2(i)\right)$$

Solving for $\dot{\omega} = 0$ gives the value $i = 63.4^\circ$, that meaning the orbit has its apogee stuck in place at the same latitude. These kind of orbits are semi-synchronous, that meaning its orbital period is chosen to be half a sidereal day, granting it a repeating ground track. The semi-major axis is 26562 km, its eccentricity is around 0.73 and the argument of perigee can be freely chosen as to place the apogee over the northern latitudes to 270° .

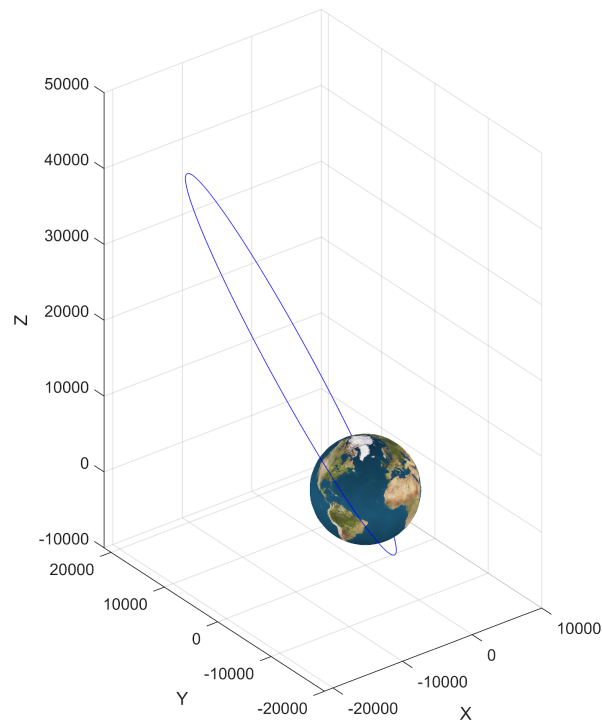


FIGURE 3. Molniya orbit trajectory

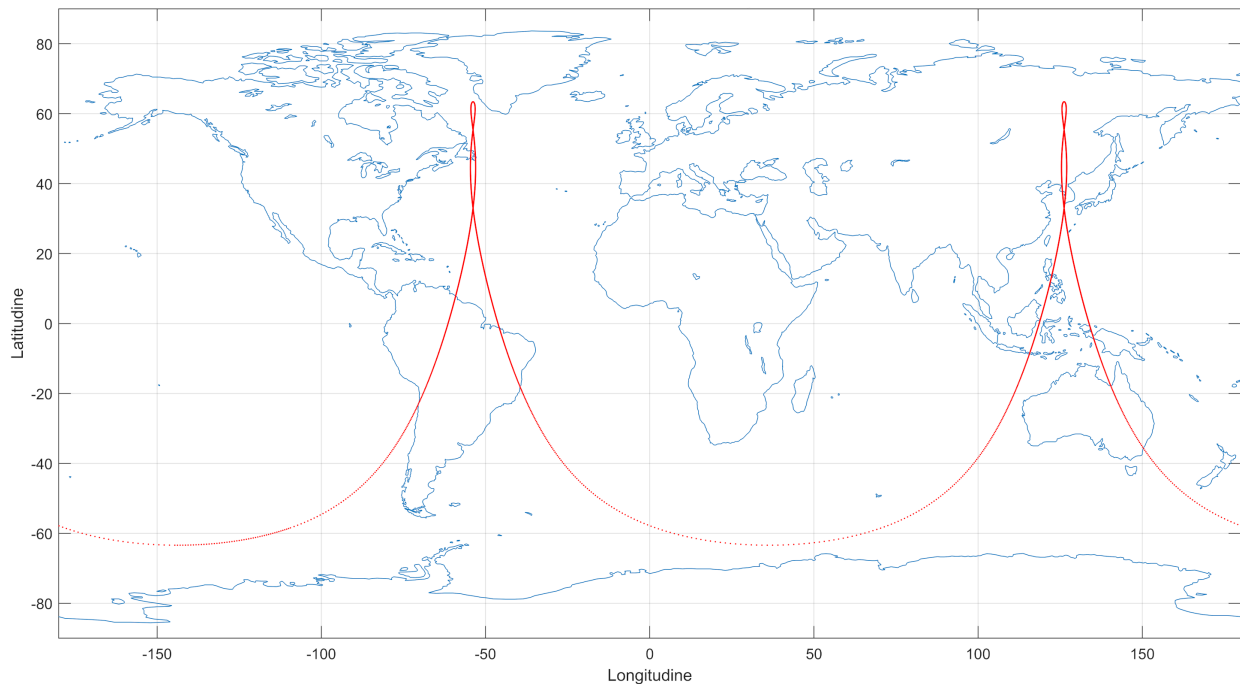


FIGURE 4. Molniya orbit ground track

- **Geosynchronous orbit**

The *Geosynchronous orbit* is a high orbit (it's a HEO, which guarantee an orbital period of 1 sidereal day (86164 s), thus allowing a spacecraft to match Earth's rotation.

If a geosynchronous orbit is circular and equatorial then it is also a *Geostationary orbit*, and will maintain the same position relative to the Earth's surface.

In the more general case, when the orbit has some inclination and/or eccentricity, the satellite would appear to describe a more or less distorted figure-eight in the sky, also called an analemma and would rest

above the same spots of the Earth's surface once per sidereal day.

Having a semi-major axis of 42164 km, this orbit is valuable for monitoring weather, communications and surveillance. A satellite in geosynchronous orbit can see one spot of the planet almost all of the time. For Earth observation, this allows the satellite to look at how much a region changes over months or years.

A further form of geosynchronous orbit is the theoretical space elevator. When one end is attached to the ground, for altitudes below the geostationary belt the elevator maintains a shorter orbital period than by gravity alone.

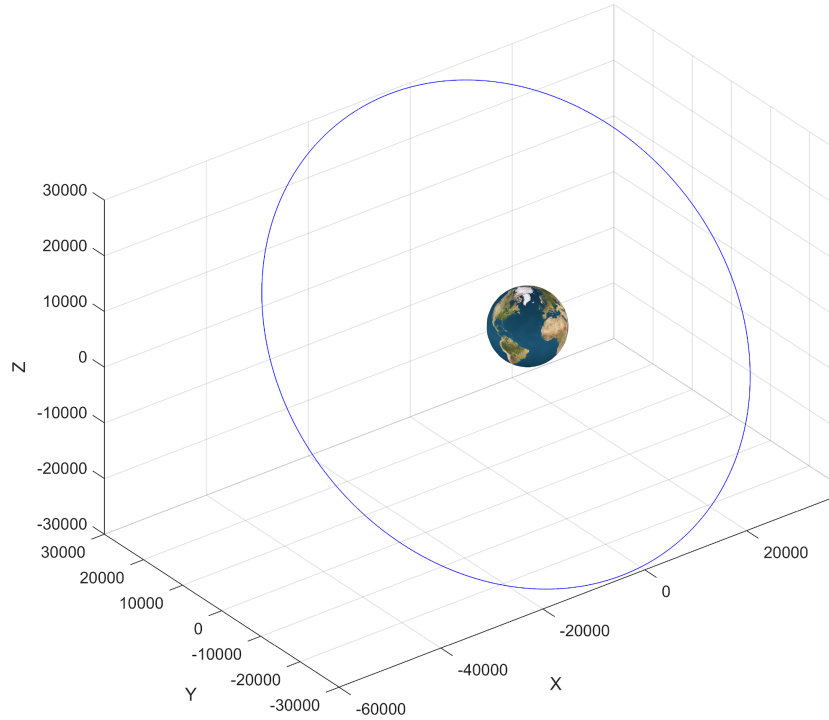


FIGURE 5. Geosynchronous orbit trajectory

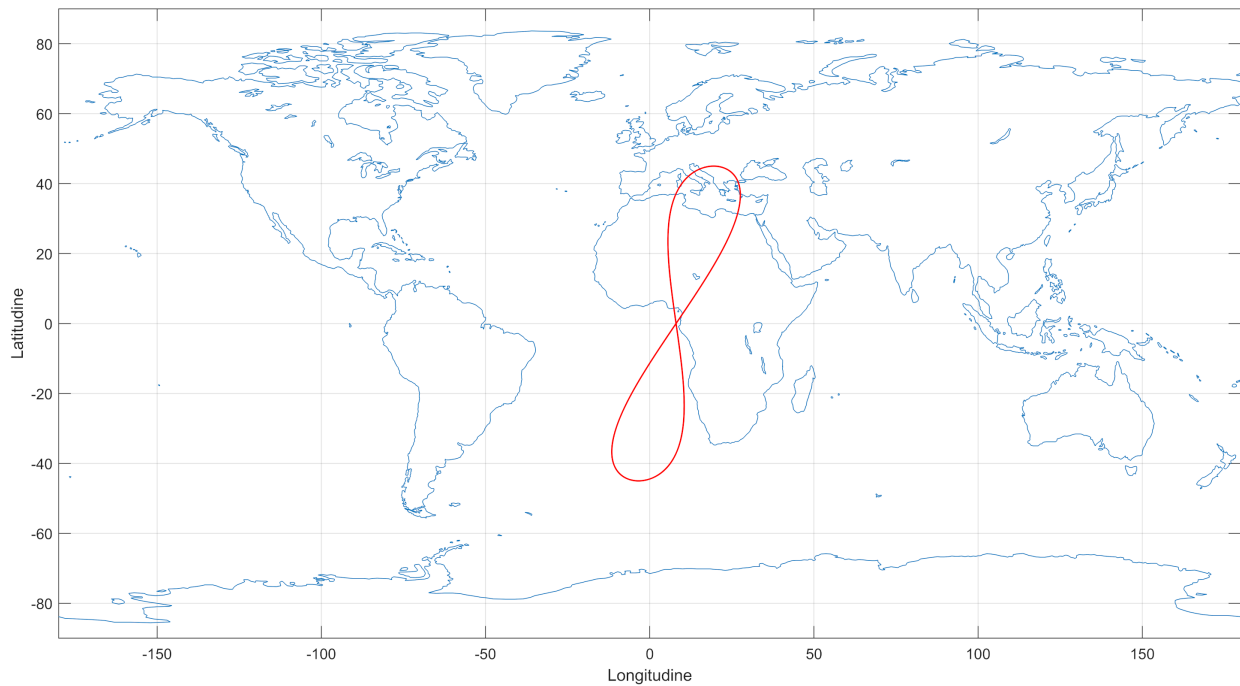


FIGURE 6. Geosynchronous orbit ground track

2 Part 2

2.1 Problem

Using the Molniya orbit as reference, estimate analytically and verify numerically the time spent by the spacecraft with a latitude above 45° N.

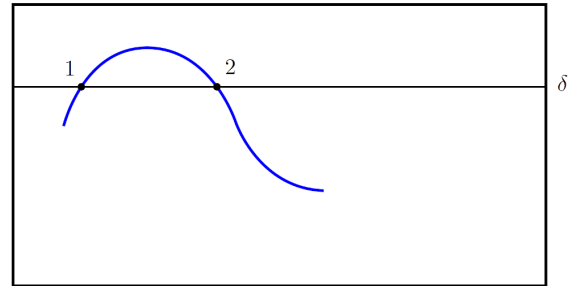
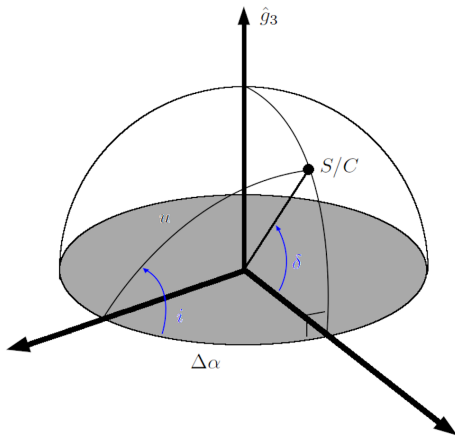
2.2 Estimation of the time spent over 45 degrees

Using the spherical law of sines on the triangle shown in the figure (where $\delta = 45^\circ$) we can correctly estimate the corresponding true anomalies to the points 1 and 2 along the ground track, and thus look for the time in between. We notice that $u = \omega + \nu$, with ω being 270° for the Molniya orbit.

$$\begin{aligned} \frac{\sin u}{\sin \frac{\pi}{2}} &= \frac{\delta}{i} \\ u_1 &= \arcsin\left(\frac{\sin \delta}{\sin i}\right) \\ u_2 &= \pi - u_1 \end{aligned}$$

We obtain the following values

$$\begin{cases} u_1 = 52.26^\circ \\ u_2 = 127.74^\circ \end{cases} \Rightarrow \begin{cases} \nu_1 = -217.74^\circ \\ \nu_2 = -142.26^\circ \end{cases} \Rightarrow \begin{cases} E_1 = 99.30^\circ \\ E_2 = 260.70^\circ \end{cases} \Rightarrow \begin{cases} M_1 = 58.52^\circ \\ M_2 = 301.48^\circ \end{cases}$$



$$\Rightarrow \Delta t = (M_2 - M_1) \sqrt{\frac{a^3}{\mu}} = \begin{cases} 8.08 \text{ hrs along one orbit (12 hours)} \\ 16.16 \text{ hrs along two orbits, (24 hours)} \end{cases}$$

3 Part 3

3.1 Problem

Design a constellation of three Molniya spacecrafts, with period $12 \text{ h}^{\text{sid}}$, that share the same ground track and are evenly distributed. Also, at the time t_A (15/01/1995 @ 00:00:00) a spacecraft must be at the orbit apoapsis

and have a longitude equal to the one in Moscow ($55^{\circ}45'06''$ N ; $37^{\circ}37'04''$ E). Graph the trajectories of the three spacecrafts in an inertial reference system and the ground track of the constellation in the time interval $t \in [t_A - 4h, t_A + 4h]$.

3.2 Estimation of COE

We first need to establish the time interval, so we'll choose a span of 8 hours from 14/01/1995 @ 20:00:00 to 15/01/1995 @ 04:00:00. Also, we are given an orbit time, $T = \frac{T_{sid}}{2}$, from which we can estimate the semi-major axis of the resulting orbit used in the constellation. Next, we choose a height of 1200 km over the perigee, typical of Molniya orbits. From the relations below we are able to come up with the equation we need to find the different Ω .

$$t_{span} = \left[49732 - \frac{1}{6}, \dots, 49732 + \frac{1}{6} \right] \times T_{solare}$$

$$t_A = 49732 \times T_{solare}$$

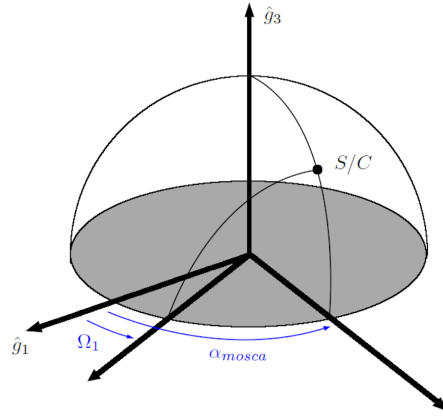
$$\alpha_{mosca}(t_A) = \alpha_{GR}(t_A) + LONG_{mosca}$$

$$\alpha_{mosca}(t_A) = \alpha_{S/C_1}(t_A)$$

$$\alpha_{S/C_1}(t_A) = \Omega_1 + \Delta\alpha(t_A)$$

$$\frac{\sin u(t_A)}{\sin \frac{\pi}{2}} = \frac{\sin \Delta\alpha(t_A)}{\sin \Psi(t_A)}$$

$$\Delta\alpha(t_A) = u(t_A) = \omega + \nu(t_A) = -\frac{\pi}{2} + \pi = \frac{\pi}{2}$$



$$a = \sqrt[3]{\frac{\mu \left(\frac{T}{2}\right)^2}{4\pi^2}}$$

$$e = 1 - \frac{R_{\oplus} + h}{a}$$

$$\Omega_1 = \alpha_{GR}(t_A) + LONG_{mosca} - \Delta\alpha(t_A)$$

$$\Omega_2 = \Omega_1 - \frac{2\pi}{3}$$

$$\Omega_3 = \Omega_1 + \frac{2\pi}{3}$$

$$\nu_1 = \pi$$

$$\nu_1 \Rightarrow E_1 \Rightarrow M_1$$

$$M_2 = M_1 - \frac{2\pi}{3}$$

$$M_3 = M_1 + \frac{2\pi}{3}$$

$$M_2, M_3 \Rightarrow E_2, E_3 \Rightarrow \nu_2, \nu_3$$

Orbit	a [km]	e	i [deg]	ω [deg]	Ω [deg]
Molniya 1	2.6562e+04	0.7147	63.4000	270.0000	61.4224
Molniya 2	2.6562e+04	0.7147	63.4000	270.0000	301.4224
Molniya 3	2.6562e+04	0.7147	63.4000	270.0000	181.4224

TABLE 3. Molniya constellation elements

3.3 Trajectory and ground track of the constellation

Finally we can draw the constellation trajectory and ground track where the orbits of the spacecrafts are shown in different colors to easily recognize them. It's noticeable as they all have the same ground track, as we wanted from the start.

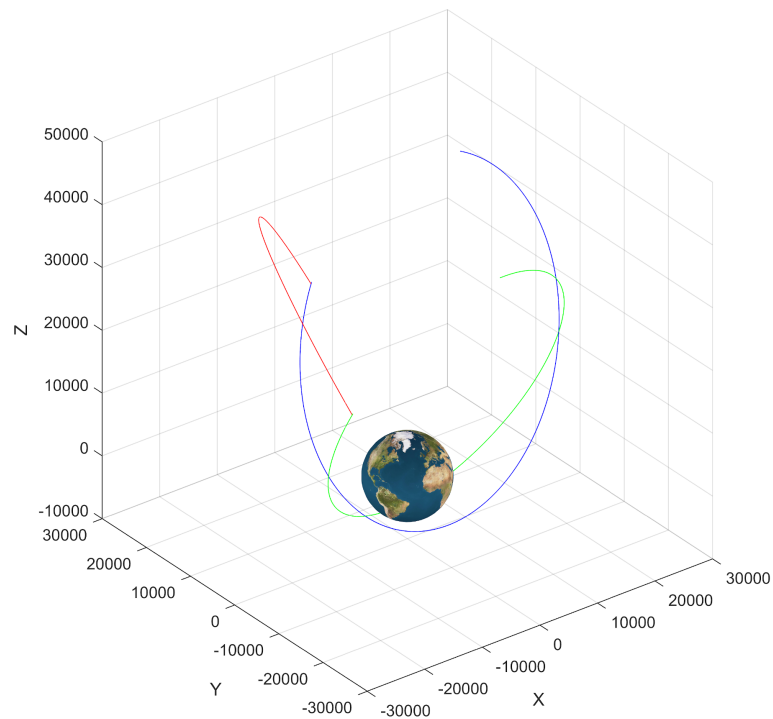


FIGURE 7. Molniya constellation trajectory

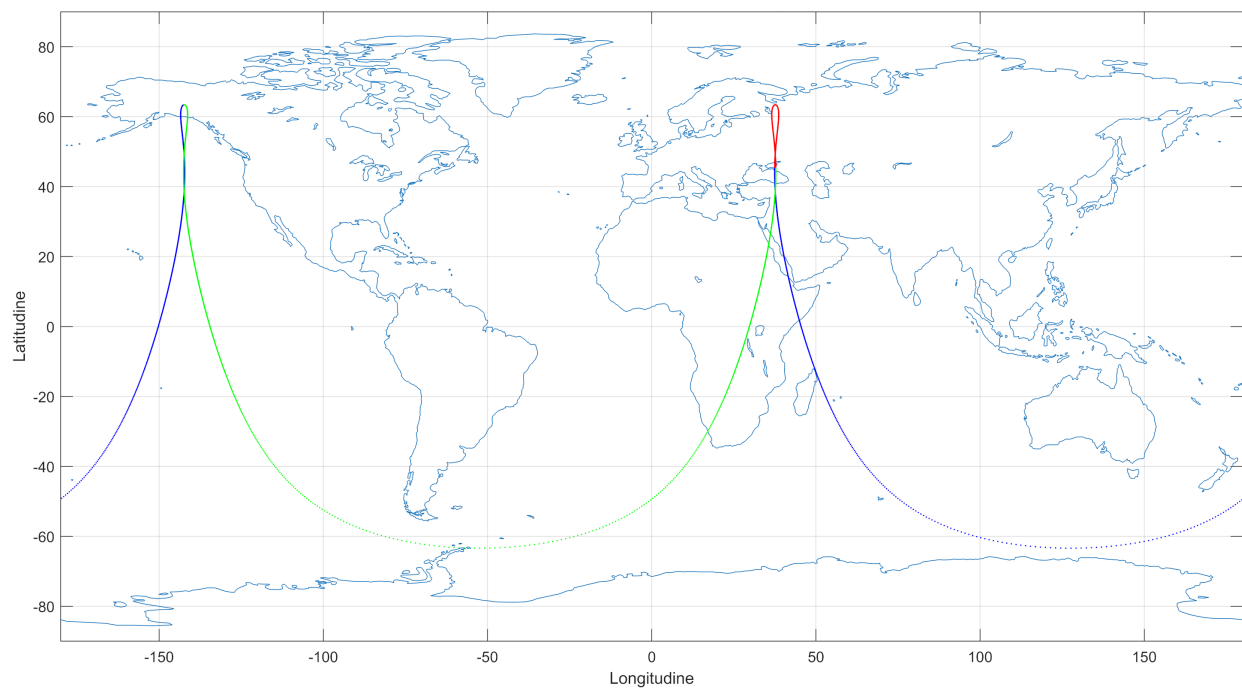


FIGURE 8. Molniya constellation ground track